School of Science, Computing and Emerging Technologies

Unit Outline

COS20028

Big Data Architecture and Application

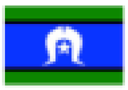
Semester May 2026

Please read this Unit Outline carefully. It includes:

PART A Unit summary

PART B Your Unit in more detail

PART C Further information



"Swinburne University of Technology recognises the historical and cultural significance of Australia's Indigenous history and the role it plays in contemporary education

Each day in Australia, we all walk on traditional Indigenous land

We therefore acknowledge the traditional custodians of the land that our Australian campuses currently occupy, the Wurundjeri people, and pay respect to Elders past and present, including those from other areas who now reside on Wurundjeri land"

PART A: Unit Summary

Unit Code(s)	COS20028
Unit Title	Big Data Architecture and Application
Duration	One Semester
Total Contact Hours	48 hours
Requisites:	
Pre-requisites	[COS10022 – Introduction to Data Science] AND [COS20007 – Object-Oriented Programming or COS30016 – Programming in Java]
Co-requisites	Nil
Concurrent pre-requisites	Nil
Anti-requisites	Nil
Assumed knowledge	Object-oriented programming skills are expected.
Credit Points	12.5 credit points
Campus/Location	Hanoi - Ho Chi Minh Campus
Mode of Delivery	Blended
Assessment Summary	Assignment 1 (Individual, 20%) Assignment 2 (Individual, 20%) Online Test 1 (Individual, 15%) Online Test 2 (Individual, 15%) Online Test 3 (Individual, 30%)

Aims

This unit introduces students to the architectures of contemporary Big Data platforms and ecosystems of tools. These technologies are the foundation of Big Data analytics, which facilitate scalable management and processing of vast quantities of data. The unit is delivered in collaboration with industry and prepares students with the essential foundations towards undertaking professional certifications.

Unit Learning Outcomes

Students who successfully complete this unit can:

1. Explain and compare the architecture of contemporary Big Data tools and platforms.
2. Demonstrate skills in working with a contemporary Big Data platform to manage large data sets.

3. Apply Big Data related technologies in developing applications to solve common problems faced by organisations.

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

This unit contributes to the development of the following Swinburne Graduate Attributes:

- GA1 Communication - Verbal communication: Assignment 1 requires a video recording to explain the design idea of your solution.
- GA2 Communication - Communicating using different media: Assignment 2 requires multiple media and tools to complete the content.
- GA5 Digital literacies– Information literacy: All assignments and tests require students to process the data with the knowledge of decomposing the structure in the data and categorising them by the extracted information.
- GA6 Digital Literacies– Technical literacy: Assignment 2, online test 2, and online test 3 require students to be familiar with different tools to be used for corresponding purposes for big data.

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

Content

- Introduction to Big Data platforms and their ecosystems
- Big Data platform programming
- Partitioners and Reducers
- Big Data acquisition and workflows
- The Data Analytics Lifecycle
- Analysis and Management of structured and unstructured Big Data
- Graduate Attribute – Communication Skills: Verbal communication
- Graduate Attribute – Communication Skills: Communicating using different media
- Graduate Attribute – Digital Literacies: Information literacy
- Graduate Attribute – Digital Literacies: Technical literacy

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- The VM is rebuilt with updated software and includes the new tool.
- Assignment activities are updated.
- Test question banks are updated.

Unit Teaching Staff

Name	Role	Location	Email / Teams	Consultation Times
Mr. Dang Hoang Anh	Unit Convenor	Hanoi	ahdang@swin.edu.au	By email appointment
Mr. Minh Hoang	Unit Convenor	Ho Chi Minh	manh@swin.edu.au	By email appointment

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
Live Online	Lectures	24 hours	2 hours	Weeks 1 to 12
On Campus	Class	24 hours	2 hours	Weeks 1 to 12
Online	Directed Online Learning and Independent Learning	12 hours	1 hour	Weeks 1 to 12
Unspecified Activities	Independent Learning	90 hours	7.5 hours	Weeks 1 to 12

Week by Week Schedule

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1	May 11	▪ **Lecture**: Overview of Hadoop. ▪ **Online materials**: Week 1 lecture slides and videos. ▪ **Lab**: Setting up and configuring the VM environment. Getting familiar with basic HDFS operation commands.	▪ Readings: Chapter 1 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Set up and access Hadoop environment. ▪ Using HDFS.
2	May 18	▪ **Lecture**: Hadoop Ecosystem: components, clusters, MapReduce. ▪ **Online materials**: Week 2 lecture slides and videos. ▪ **Lab**: Executing and understanding a MapReduce job.	▪ Readings: Chapter 2 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Running a MapReduce job.

3	May 25	▪ **Lecture**: MapReduce Programming **I**: Developing the first MapReduce program with Java and Streaming API (Python). ▪ **Online materials**: Week 3 lecture slides and videos. ▪ **Lab**: Work on **LT1** and **LT2**.	▪ Readings: Chapter 5 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Using ToolRunner and passing parameters. ▪ Using a Combiner. ▪ Word count and its variants.
4	June 01	▪ **Lecture**: MapReduce Programming **II**: Hadoop APIs. ▪ **Online materials**: Week 4 lecture slides and videos. ▪ **Lab**: ToolRunner, Combiner, and **test 1**.	▪ Readings: Chapter 6 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Testing with LocalJobRunner. ▪ Logging. ▪ ToolRunner. ▪ Combiner. ▪ **Test 1**. (Covering Week 1 – 3 materials.)
5	June 08	▪ **Lecture**: MapReduce Programming **III**: Tips and Techniques (Debugging, LocalJobRunner, Log Files, Counters, Object Reuse, Map Only Jobs). ▪ **Online materials**: Week 5 lecture slides and videos. ▪ **Lab**: Work on LocalJobRunner, Logging, and LT3 contents.	▪ Readings: Chapter 4 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Writing a Partitioner. ▪ LocalJobRunner ▪ Logging. ▪ **Assignment 1 Due.**
6	June 15	▪ **Lecture**: MapReduce Programming IV: Partitioners and reducers, Data Input and Output. ▪ **Online materials**: Week 6 lecture slides and videos. ▪ **Lab**: Work on Partitioner, Custom WritableComparable, SequenceFile, and File Compression.	▪ Readings: Chapter 3 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Creating an Inverted Index Calculating Word Cooccurrence. ▪ Implementing a custom WritableComparable. ▪ Using Sequence Files and File Compression. ▪ Partitioner, Custom WritableComparable, SequenceFiles, and File Compression.

7	June 22	▪ **Lecture**: MapReduce Programming V: Common MapReduce algorithm. Introduction to Kafka. ▪ **Online materials**: Week 7 lecture slides and videos. ▪ **Lab**: Work on Inverted Index, and Word Co occurrence, and the **test 2**.	▪ Readings: Chapter 8 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Importing data with Sqoop. ▪ Running an Oozie workflow. ▪ Exploring a Secondary Sort example. ▪ Running Inverted Inex ▪ Running Word Co-occurence ▪ **Test 2.** (Covering Week 4 – 6 materials.)
8	June 29	▪ **Lecture**: Data Processing and Analysis I: Sqoop, Flume, Oozie, Kafka. ▪ **Online materials**: Week 8 lecture slides and videos. ▪ **Lab**: Secondary sort, Sqoop data import, Kafka application, and LT4.	▪ Readings: Chapter 15 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Data ingest with Hadoop tools. ▪ Exploring a Secondary Sort Example. ▪ Importing Data with Sqoop. ▪ Kafka exercise.
Summary Holiday – No classes from 06 – 12 July, inclusively.			
9	July 13	▪ **Lecture**: Data Processing and Analysis II: Apache Pig and ETL. ▪ **Online materials**: Week 9 lecture slides and videos. ▪ **Lab**: Data ingestion, Pig for ETL, Analysing AD campaign, and LT5 contents.	▪ Readings: Chapter 16 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Analysing disparate datasets with Pig. ▪ Analysing Ad campaign data with Pig.
10	July 20	▪ **Lecture**: Data Processing and Analysis III: Pig. ▪ **Online materials**: Week 10 lecture slides and videos. ▪ **Lab**: Analysing disparate data with Pig and LT6.	▪ Readings: Chapter 17 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Apache Pig for analysing disparate data.
11	July 27	▪ **Lecture**: Data Processing and Analysis IV: Hive. ▪ **Online materials**: Week 11 lecture slides and videos. ▪ **Lab**: HiveQL and LT7 content.	▪ Readings: Readings: Chapter 12 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Sentiment analysis with Hive. ▪ Data transformation with Hive. ▪ Alter and drop a table. ▪ **Assignment 2 Due.**

12	Aug 03	▪ **Lecture**: Contemporary Issues in Big Data Analytics. ▪ **Online materials**: Articles on Blogs. ▪ **Lab**: **Test 3**.	▪ Readings: Readings: Chapter 22 of "Hadoop: The Definitive Guide, 4th Edition." ▪ **Test 3.** (Covering week 7 – 11 materials.)

Assessment

a) Assessment overview

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
Assignment 1	Individual	20%	2	1, 5	Week 05
Assignment 2	Individual	20%	1, 2, 3	2, 5, 6	Week 11
Test 1	Individual	15%	1, 2, 3	5	Week 04 Class
Test 2	Individual	15%	1, 2, 3	5, 6	Week 07 Class
Test 3	Individual	30%	1, 2, 3	5, 6	Week 12 Class

Assessment Requirements	Details
b) Use of generative AI (genAI) in this unit	You are not allowed to use any AI tool to produce contents in your assignments or to answer the questions in the test. Moreover, uploading any of the material from this unit to any of the generative AI (genAI) tool such as ChatGPT, Llama, etc. are all prohibited. However, there are some scenarios you can use the generative AI tool The valid use of genAI in this unit is as follows: 1. Assist your learning in producing practice questions for you to prepare the test. 2. Assist you to correct gramma issues in the report. Nevertheless, any use of genAI must be acknowledged, with prompts and outputs included in an appendix.

c) Hurdle requirements	The basic requirement for passing this unit is to accumulate at least 50 marks in the unit. Reattempt submission is not accepted in this unit. We have the test hosted in your registered session. If you are late for the test, you can only take the remaining time to do the test before the session is closed, as other units are scheduled to use the computer room. If you miss the test, a 0 mark will be assigned to the corresponding test. No alternative test or assignment can be arranged.
d) Final assessment period	There is no final exam in this unit. All tests in this unit are closed book and organised in your registered lab sessions. You will be expected to be available for your lab session to take the tests.
e) Submission requirements	Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin. Please ensure you keep a copy of all assessments that are submitted. All assignments in this unit are provided with instructions. You are expected to follow the given sequence to answer the questions and find answers. Some questions need to be answered based on the answers you found in the previous questions. If you are not following the instructed sequence, you may get incorrect answers and get a deduction from your marks. Turnitin is built into the Canvas submission system. You are expected to submit your assignments after passing the similarity check. The similarity value must not exceed 30% except for the parts coming from the template or some short answers. If the similarity of your submission is too high, your tutor will contact you and ask you to resubmit an updated version within the given time. The assessment cover sheet should be submitted with the assignment report as a single file and pass the Turnitin check. Some assessments may come with the interview in the lab session. If you cannot pass the interview in the registered lab session, your submission may receive 0 marks or be deducted as you failed to explain how the answers are obtained in a reasonable way. Codes are generally required for each assignment to

	be submitted independently. If your code is identified as sourced from other students' submissions, all involved submissions will receive a deduction from the marks. In cases where a hard copy submission is required, an Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the Submitting work webpage or www.swinburne.edu.au/studentforms/
f) Extensions and late submissions	Extensions – Extensions can only be issued according to a valid medical certificate issued by authorised GP in Melbourne. A valid medical certificate should reveal information regarding the patient and the impact dates unfit to work or study. The convenor will assess the overlap between the impact dates and days before the submission deadline to decide whether to approve the extension application and the days of extension can be given. A medical certificate can only be used once in each assessment. If you suffer from illness longer than the medical certificate states, you need to see the GP and get another valid medical certificate to apply for further extension. If the extension goes beyond the date of submitting the final result to the university, a special exam covers all contents in this unit is required for the student to take. The mark for the affected assignment will be the average mark achieved in the corresponding assignment and the special exam. Late Submissions – Unless an extension has been approved, late submissions will result in a penalty. You will be penalised 10% of the full mark for each day the task is late, up to a maximum of 5 days. After 5 days, a zero result will be recorded. The penalty is calculated directly by the Canvas system. We follow whatever the system produces in the penalty result.
g) Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit are: - The prompt used to get the responses. - The responses from the AI tool.

	Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/
h) Groupwork guidelines	There is no group work in this unit.

Required Textbook(s)

The required textbook(s) can be purchased from bookshops and may be available through the Swinburne Library.

- Hadoop: the definitive guide, White, Tom (Tom E.) author., 4th ed.., Sebastopol, California: O'Reilly, 2015.
- Hadoop operations, Sammer, Eric., Loukides, Michael Kosta.; Nash, Courtney.; Romano, Robert (Illustrator), illustrator., First edition., Sebastopol, CA: O'Reilly, 2012

Recommended Reading Materials

Swinburne Library has a large collection of resources. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

- Programming pig: dataflow scripting with hadoop, Gates, Alan, author.; Dai, Daniel, 2nd edition., Beijing, China: O'Reilly, 2017

PART C: FURTHER INFORMATION

For further information on any of these topics, refer to Swinburne's Student webpage http://www.swinburne.edu.au/student/

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The Student Charter describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the Student Academic Misconduct Regulations, Student General Misconduct Regulations and the People, Culture and Integrity Policy. Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or https://swinburne.instructure.com/ Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence; contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and non compliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site • creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own

without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgment
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see https://www.swinburne.edu.au/student-login/academic-integrity/

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the Current students web page.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See https://www.swinburne.edu.au/life-at-swinburne/student support-services/special-consideration-assistance/

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact Accessibility Services to discuss further.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See https://www.swinburne.edu.au/corporate/feedback/

Advocacy

Should you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Academic Student Support Officer at Swinburne Student Life.

For an appointment, please call 19006412 or email swin@fe.edu.vn .For more information, please see https://portal.swin.edu.vn